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COURSE CODE: ECA0905
COURSE NAME: DIGITAL SIGNAL PROCESSING

PRACTICAL LAB RESULTS

EXP 3

Program:

```
clc; clear; close all;

% Specifications

Fs = 2000;

Rp = 1; % Passband ripple (dB)

Rs = 40; % Stopband attenuation (dB)

% Normalized passband and stopband edge frequencies
Wp = [300 600] / (Fs/2);
Ws = [200 700] / (Fs/2);

% --- Butterworth Filter Design ---
[n1, W1] = buttord(Wp, Ws, Rp, Rs);
[b1, a1] = butter(n1, W1); % Removed 'bandpass' string

% --- Chebyshev Type-I Filter Design ---
[n2, W2] = cheb1ord(Wp, Ws, Rp, Rs);
[b2, a2] = cheby1(n2, Rp, W2); % Removed 'bandpass' string

% --- 1. Magnitude & Phase Response ---
figure('Name', 'Magnitude & Phase Responses');
subplot(2, 1, 1);
freqz(b1, a1, 1024, Fs);
title('Butterworth Filter Response');
subplot(2, 1, 2);
freqz(b2, a2, 1024, Fs);
title('Chebyshev Type-I Filter Response');

% --- 2. Group Delay Comparison ---
figure('Name', 'Group Delay Comparison');
[gd1, w1] = groupdelay(b1, a1, 1024, Fs);
[gd2, w2] = groupdelay(b2, a2, 1024, Fs);
plot(w1, gd1, 'b-', 'LineWidth', 1.5); hold on;
plot(w2, gd2, 'r-', 'LineWidth', 1.5); hold off;
grid on;
```

```

%Frequency (Hz);
%Group Delay (samples);
title('Group Delay Comparison');
legend('Butterworth', 'Chebyshev-I');

% --- 3. Pole-Zero Plot ---

figure('Name', 'Pole-Zero Maps');

subplot(1, 2, 1); plot(b1, a1); title('Butterworth Pole-Zero Map');
subplot(1, 2, 2); plot(b2, a2); title('Chebyshev-I Pole-Zero Map');

% --- 4. Impulse Response ---

figure('Name', 'Impulse Responses');

subplot(2, 1, 1); plotz(b1, a1, [], Fs); title('Butterworth Impulse Response');
subplot(2, 1, 2); plotz(b2, a2, [], Fs); title('Chebyshev-I Impulse Response');

% --- 5. Pole Magnitude ---

figure('Name', 'Pole Magnitudes');

subplot(1, 2, 1); stem(abs(roots(a1)), 'filled'); title('Butterworth Pole Magnitudes'); grid on;
subplot(1, 2, 2); stem(abs(roots(a2)), 'filled'); title('Chebyshev-I Pole Magnitudes'); grid on;

% --- Display Calculated Filter Orders ---

fprintf('Butterworth Filter Order (Bandpass): %d\n', n1*2); % n1 is order of prototype
fprintf('Chebyshev-I Filter Order (Bandpass): %d\n', n2*2);

```

Outputs:

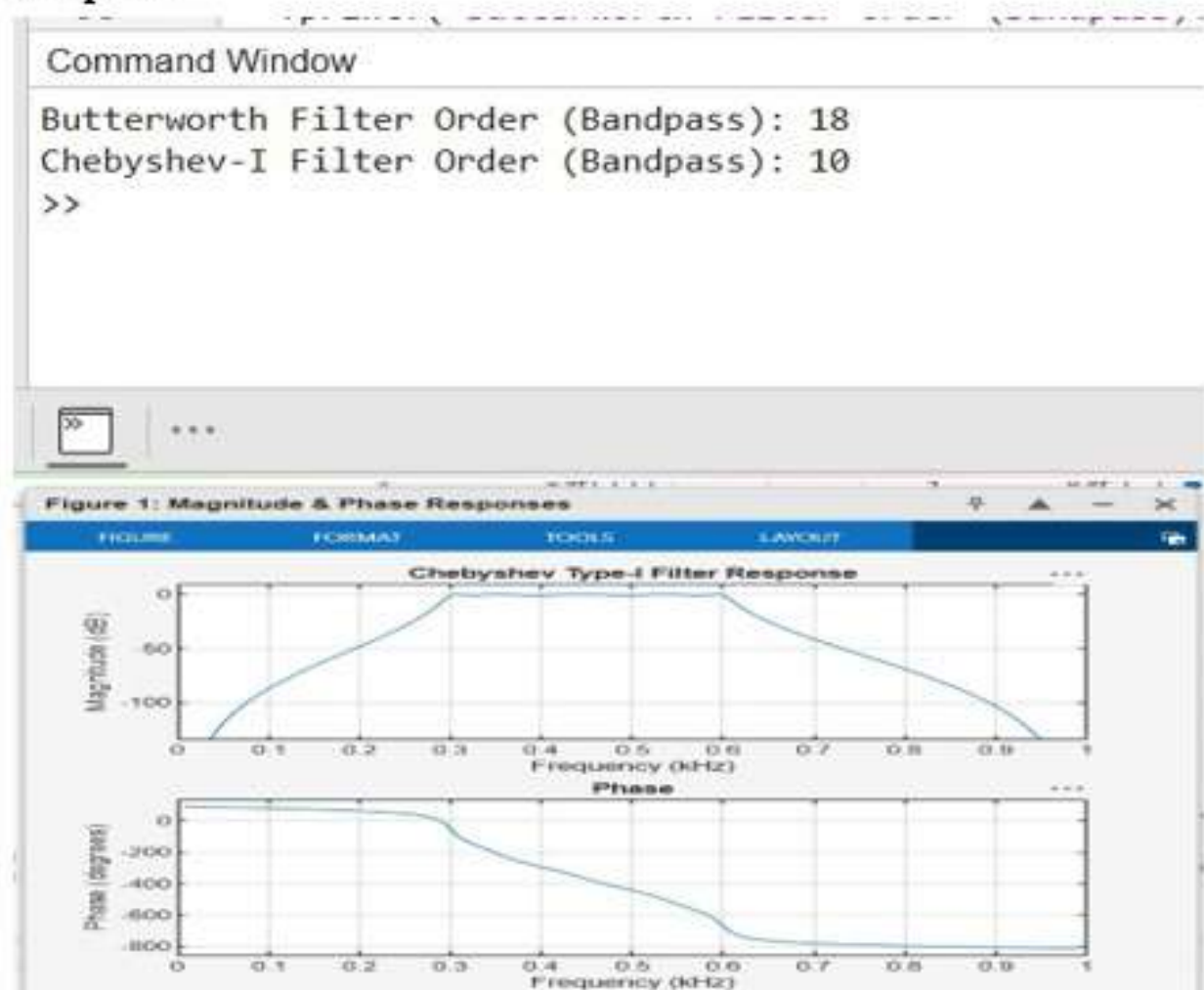


Figure 2: Group Delay Comparison



Figure 3: Pole-Zero Maps

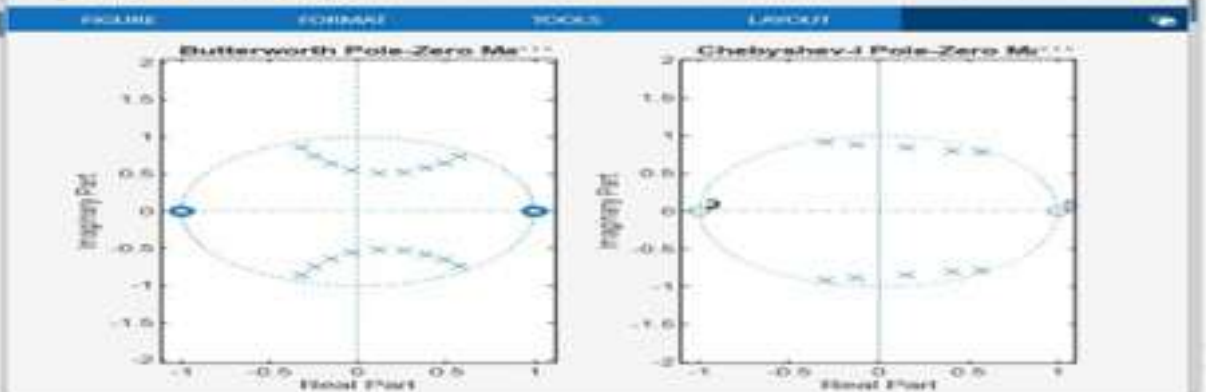


Figure 4: Impulse Responses

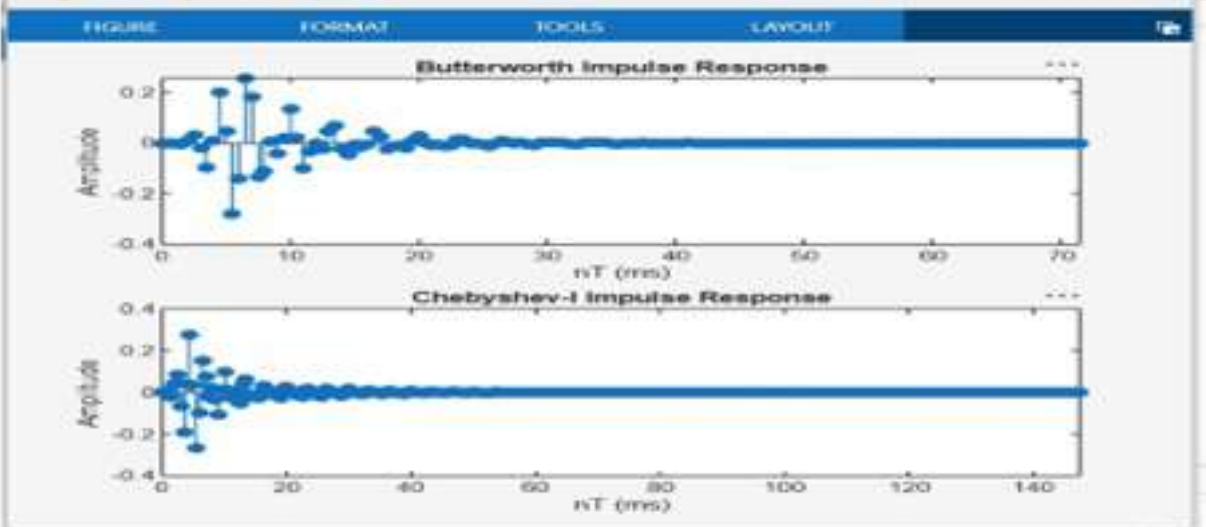


Figure 5: Pole Magnitudes

